message. ***HMAC*** (hash-based MAC) is a widely used secure MAC based on a cryptographic hash function. For example, ***HMAC-SHA-256*** is HMAC using the SHA-256 hash function.

The output of the HMAC function is a short authentication tag that can be appended to the token as shown in figure 5.8. An attacker without access to the secret key can't calculate the correct tag for a token, and the tag will change if even a single bit of the token ID is altered, preventing them from tampering with a token or faking new ones.

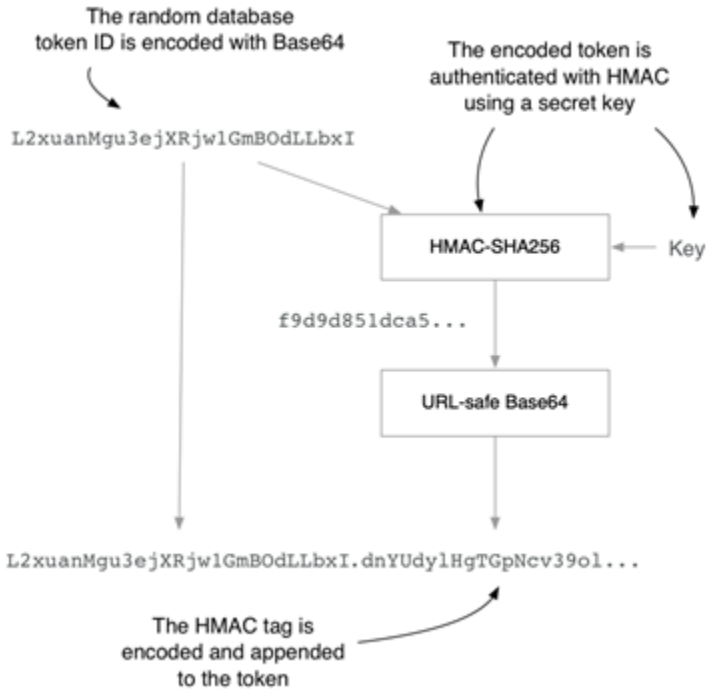


Figure 5.8 A token can be protected against theft and forgery by computing a HMAC authentication tag using a secret key. The token returned from the database is passed to the HMAC-SHA256 function along with the secret key. The output authentication